PRACTICAL EXERCISE REPORT

CLEANING AND ANALYZING HOTEL BOOKING DATA

Group 06 · Course: Data Analysis with Python · Instructor: PhD. Trần Hùng

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1. Objectives & Data sources

Objective. This report describes bookings at two hotels: cancellation rates, seasonality, lead time, guest country, market segment, deposit type, room price, length of stay and room changes. Before the analysis, we cleaned a data file that contains deliberate errors and recorded every cleaning decision. All numbers come from the notebook hotel_analysis.ipynb after a full re-run.

Unit of analysis. One booking = one booking_id. The input file has 10,325 rows. After cleaning, **10,000 bookings** remain. Every rate in this report states its denominator and filter.

Data source. Antonio, N., de Almeida, A., & Nunes, L. (2019). Hotel booking demand datasets. Data in Brief, 22, 41-49 (CC BY 4.0 licence). The course file hotel_bookings_dirty.csv is a random sample of the original 119,390 bookings, with formatting errors, missing values, contradictions and duplicate records added on purpose. It covers two hotels in Portugal: City Hotel (Lisbon) and Resort Hotel (Algarve). Arrival dates run from 1 July 2015 to 31 August 2017; the data cut-off is 14 September 2017 (the latest status update).

Scope. Ten descriptive questions: data quality before and after cleaning, cancellations by hotel, seasonality in 2016, lead time, top 10 countries, market segment, deposit type, average daily rate (ADR), length of stay and room changes. The data do **not** include actual revenue, costs, cancellation reasons, guest reviews or personal information. Therefore, the report makes no claims about why bookings were canceled, and it does not treat adr as revenue

2. Data quality & Processing

2.1 Overview before and after cleaning

Indicator	Before (base: 10,325 rows)	After (base: 10,000 bookings)	Note
Rows	10,325	10,000	After: one row per booking_id
Distinct booking_id values	10,145	10,000	Before: not yet stripped/upper-cased (bk046154 ≠ BK046154)
Exact duplicate rows (23 columns)	180	0	181 after standardising the key; removed in B1
Repeated booking_id (after standardising)	325	0	181 removed (B1) + 144 merged (B8)
Cells with a value that breaks the rules	3,955	0	Unknown labels, wrong format, out of range
Missing cells (23 columns)	311	352	Higher because invalid values became missing
Missing rate of adr	0.24%	0.36%	11 negative or > 1,000 EUR prices → missing
Missing rate of arrival_date	0.24%	0.49%	24 non-existent or out-of-range dates → missing
Bookings labelled 'City Hotel'	6,763	6,702	Before: exact spelling only, duplicates included
Cancellation rate	37.24%	37.41%	Before: only 10,120 rows coded '0'/'1', duplicates included
Maximum ADR (EUR per night)	5,400	392	Prices above 1,000 EUR are invalid (Section 5.1)

Table 1. *Data quality indicators before and after cleaning. Source: outputs/tables/q01_before_after.csv, outputs/quality_report.csv*

The two columns use different denominators: input rows (before) and bookings (after). The automated checks in the notebook (keys, labels, value ranges, derived columns, monthly table, lineage and log) passed **51 of 51** items. The cleaning log has 51 rules, and every count in it was computed by code. Besides the three decisions below, we standardized the labels of seven categorical columns, converted yes/no and True/False

flags to 0/1, and recomputed is_canceled in 40 rows that contradicted the final reservation status (Section 5.3 of the brief).

2.2 Three key cleaning decisions

Decision 1: Standardise the key first, then remove duplicates and merge. 215 booking_id values had extra spaces or lower-case letters (e.g. R00096, R00110); we stripped and upper-cased them. We then removed 181 rows that were exact duplicates on all 23 business columns (e.g. R00608, R00887) and merged the remaining 144 rows that shared a booking_id, taking the first non-missing value in row_id order. Fifteen bookings had real conflicts in adr (8) or lead_time (7), all small (e.g. lead_time 236 vs 239); we kept the value from the row with the smallest row_id. We did not merge 824 rows that were identical except for booking_id, because they may be different rooms booked by the same group. Impact: every rate uses 10,000 bookings as its base; without this step, bookings and cancellations would be counted twice.

Decision 2: Read dates explicitly, don't guess. arrival_date appeared in four formats, and 1,931 dates were not in ISO form (e.g. R00005, "October 29, 2016"). We parsed each format in turn with errors="coerce" and always read slash dates as day/month/year. Sixteen dates that did not exist or could not be read (not_a_date, 31/02/2016) and eight dates outside July 2015-August 2017 were set to missing instead of being moved to the nearest valid date. Twenty-six dates found in the reservation_status column were moved to reservation_status_date, but only where that cell was empty. Impact: the seasonality analysis (Question 3) and the monthly table have complete arrival months; the 49 bookings without an arrival date are excluded only from monthly calculations.

Decision 3: Clean adr but keep rare valid values. We removed the €/EUR symbol from 39 prices and converted 79 prices that used a decimal comma (e.g. R00432), so 95,50 became 95.5, not 9550. Eleven prices that were negative or above 1,000 EUR (the highest was 5,400) were set to missing. Rare values inside the valid range, such as adr = 0, zero-night stays and lead times of 600 days or more (146 rows) were kept. Impact: the maximum ADR fell from 5,400 to 392 EUR; in Question 8, the 120 non-canceled bookings with adr = 0 are excluded from price statistics and reported separately.

2.3 Why missing values were kept

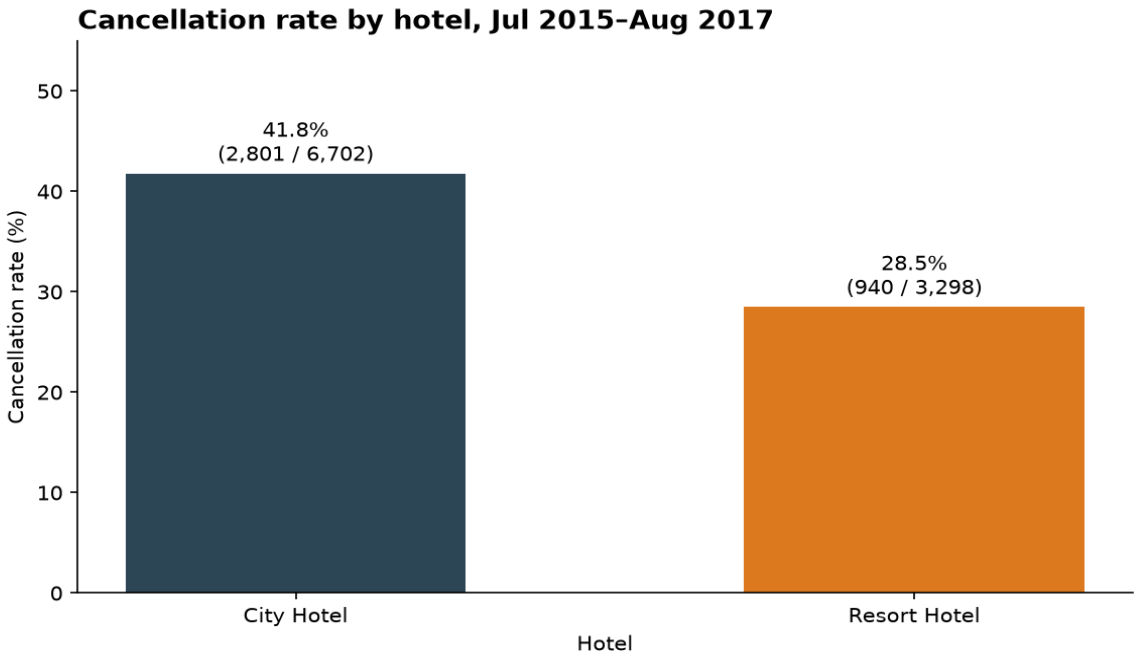
Following the rules of the brief, values that could not be corrected with the available evidence were left as missing (pd.NA). We did not fill them with the mean or the most common value, because that would create data that do not exist and pull the statistics towards the majority. We also did not drop all rows with missing cells: a booking without a country is still valid when computing the cancellation rate by hotel. Each question

excludes only the bookings that lack the column it needs and reports how many were excluded (e.g. 107 bookings without a country in Question 5 and 31 without lead_time in Question 4).

The missing rates of some columns **rose** after cleaning: adr from 0.24% to 0.36%, lead_time from 0.10% to 0.31% and arrival_date from 0.24% to 0.49% because invalid values (negative prices, "unknown", 31/02) were turned into missing values. This does not mean the cleaning was poor: the earlier data only looked complete. Other columns had fewer missing values after rows with the same booking_id were merged. For example, all 24 empty meal cells belonged to bookings with duplicate rows and were filled from those rows. reservation_status is missing for 25 bookings (dates in the wrong column), but is_canceled is still known, so these bookings stay in the cancellation denominator.

3. Discovery from data

3.1 Insight 1: City Hotel exhibits a significantly higher attrition risk than Resort Hotel across temporal periods

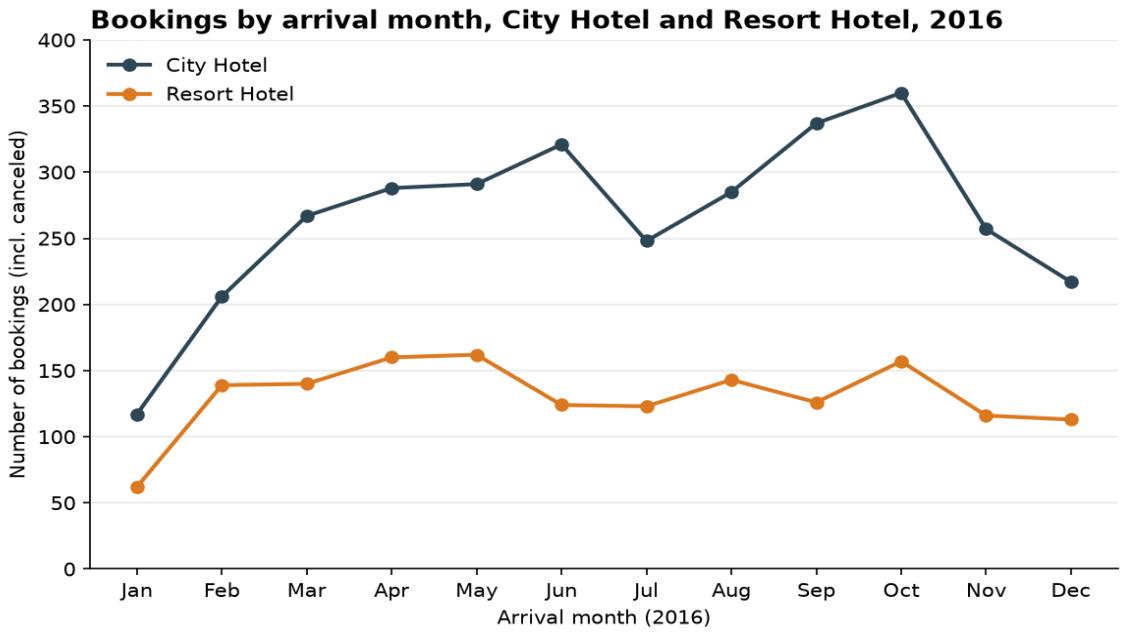


n = 10,000 bookings (No-Show counted as canceled); none excluded. Labels: rate (canceled / bookings). Source: outputs/tables/q02_hotel_cancel.csv

Figure 1. *Cancellation rate by hotel in Jul 2015 - Aug 2017, n = 10,000 bookings. Source: outputs/tables/q02_hotel_cancel.csv*

Out of 6,702 valid bookings at City Hotel, 2,801 were canceled (41.79%), representing a margin of 13.29 percentage points above Resort Hotel (28.50% of 3,298 bookings), or

approximately a 1.47-fold relative increase. Within this empirical sample, City Hotel demonstrates a heightened baseline attrition risk requiring rigorous operational monitoring. Nevertheless, given the intrinsic structural differences in property typology, geographical positioning, and distribution channel architecture, incorporating qualitative data on cancellation rationales is imperative to substantively explain this aggregate variance.



n = 4,759 bookings with a 2016 arrival date. Not compared with 2015 or 2017 (incomplete years). Source: outputs/tables/q03_monthly_arrivals_2016.csv

Figure 2. *Bookings by arrival month in 2016, n = 10,000 bookings. Source: q03_monthly_arrivals_2016.csv*

This attrition discrepancy is not confined to isolated seasonal anomalies: throughout 2016, City Hotel's cancellation rate consistently exceeded that of Resort Hotel in 9 out of 12 observed months. The only exceptions occurred during the peak summer window (July and August, where Resort Hotel peaked at 41.26% versus City Hotel's 34.39%) and February (marginal parity). This temporal consistency confirms that the higher attrition risk at City Hotel is a structural, year-round phenomenon rather than a transient seasonal artifact.

3.2 Insight 2: Extended lead times demonstrate a strong positive correlation with observed cancellation rates

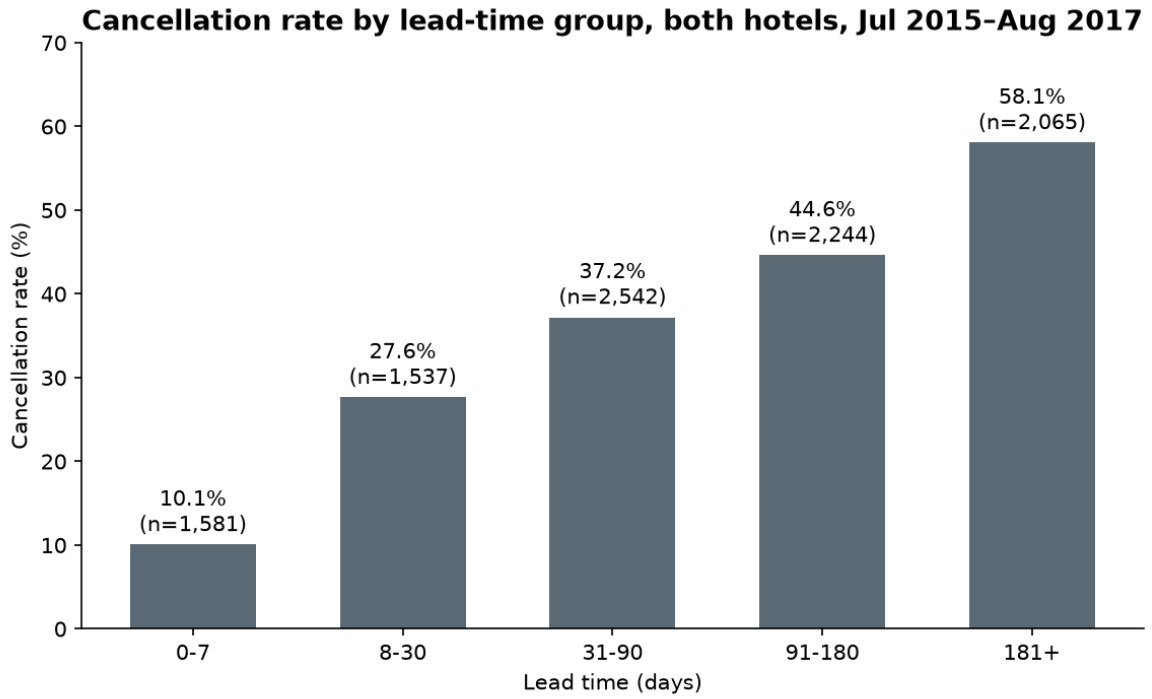
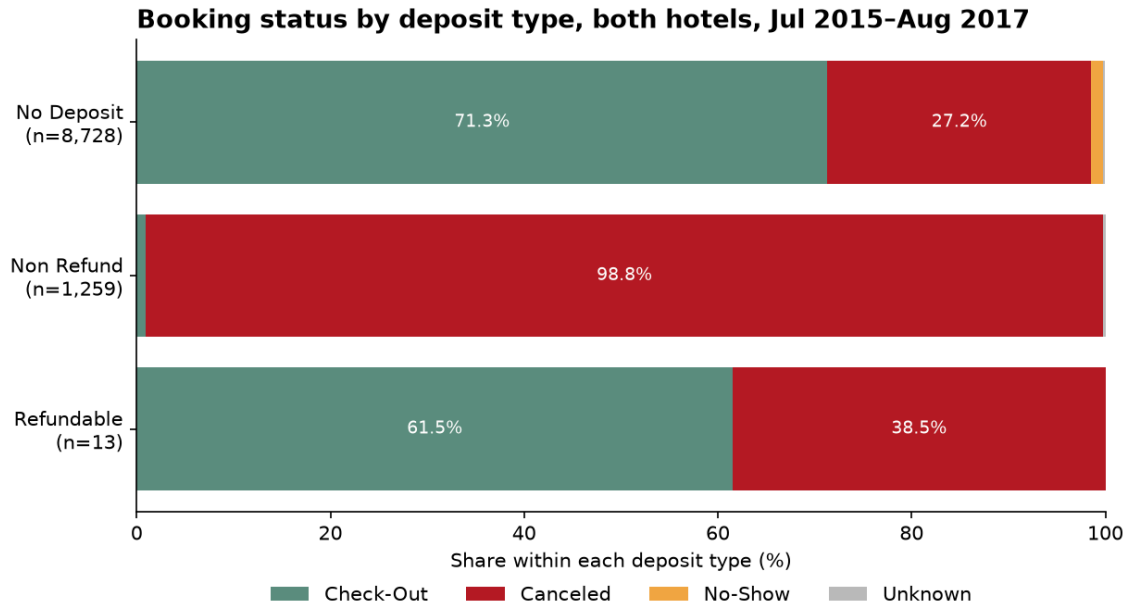


Figure 3. *Cancellation rate by lead time group. n = 9,969 bookings (excluding 31 missings). Source: outputs/tables/q04_leadtime_cancel.csv*

Among the 9,969 bookings with valid lead time metrics, the cancellation rate exhibits a monotonic increase across successive temporal cohorts: from 10.06% in the 0-7 days segment (n = 1,581) to 37.21% in the 31-90 days segment, culminating at 58.06% in the 181+ days segment (n = 2,065) - yielding an absolute variance of 48 percentage points. This upward trajectory is prevalent in both City Hotel (12.54% → 65.44%) and Resort Hotel (6.76% → 41.15%), indicating a systemic pattern rather than a property-specific anomaly. While lead time functions as a robust descriptive proxy for attrition risk, this relationship remains strictly correlational. Notably, the 181+ days cohort is structurally disproportionate, comprising 37.7% Group reservations and 31.3% Non-Refundable bookings, which act as confounding variables.

3.3 Insight 3: Non-Refundable bookings exhibit a near-total cancellation rate; however, causal attribution to deposit policy remains unsubstantiated



n = 10,000 bookings; Unknown = missing status (kept in the denominator). Refundable has a very small sample. No causal claim. Source: outputs/tables/q07_deposit_cancel.csv

Figure 4. *Reservation status by deposit type. n = 9,987 bookings (Non Refund and No Deposit). Source: outputs/tables/q07_deposit_cancel.csv*

Of the 1,259 Non-Refundable bookings, 1,244 defaulted to a Canceled status, yielding an attrition rate of 99.05%. This metric stands 70.5 percentage points higher than the No Deposit cohort (28.52% of 8,728 bookings). Crucially, the Non-Refundable subset is highly skewed: it is predominantly concentrated in City Hotel (1,106 bookings), the Groups segment (813), and Offline TA/TO distribution channels (410), featuring a median lead time of 186 days (versus 57 days for No Deposit). Consequently, this severe discrepancy is highly likely an artifact of systemic data recording methodologies or specific macro-segment behaviors rather than a direct consequence of the financial penalty. Definitive conclusions necessitate external validation regarding the actual operational deposit enforcement.

3.4 Insight 4: Domestic guests constitute a plurality of the sample and demonstrate an anomalously high cancellation probability



Share of n = 10,000 bookings (107 with missing country). Country of canceled bookings may be less accurate. Source: outputs/tables/q05_top10_country.csv

***Figure 5**. Top 10 countries by booking volume and cancellation rate. n = 10,000 bookings. Source: outputs/tables/q05_top10_country.csv*

Within the 10,000 finalized bookings, guests registered under Portugal (PRT) account for 4,005 records (40.05%), exceeding the secondary demographic (GBR, 9.99%) by a factor of four. The cancellation rate for the PRT demographic is recorded at 57.68%, outpacing GBR by 38 percentage points and representing a 2.4-fold escalation compared to the aggregate non-PRT baseline (23.88% of 5,888 bookings). While a predominant domestic market share is geographically congruent for Portuguese properties, methodological constraints must be acknowledged. As stipulated by the primary dataset authors, nationality attribution is predominantly finalized at check-in. Ergo, premature cancellations likely default to the 'PRT' locus in the property management system, introducing significant validity risks into geopolitical cancellation metrics.

3.5 Insight 5: Significant structural variances exist between the properties regarding length of stay and pricing distribution

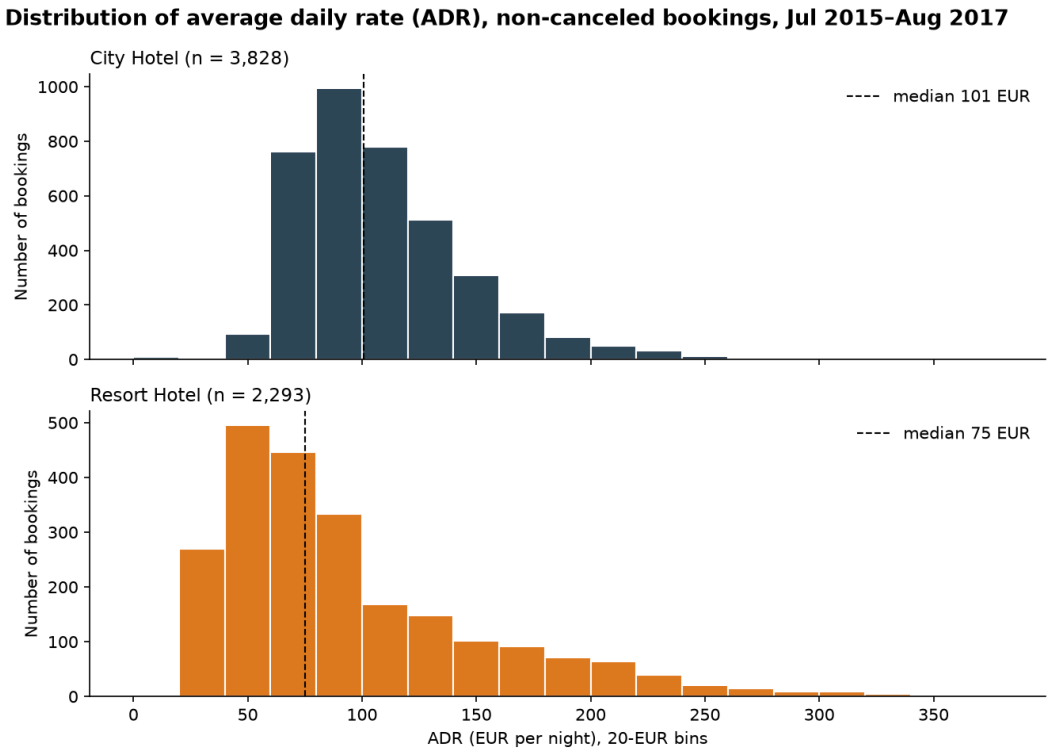
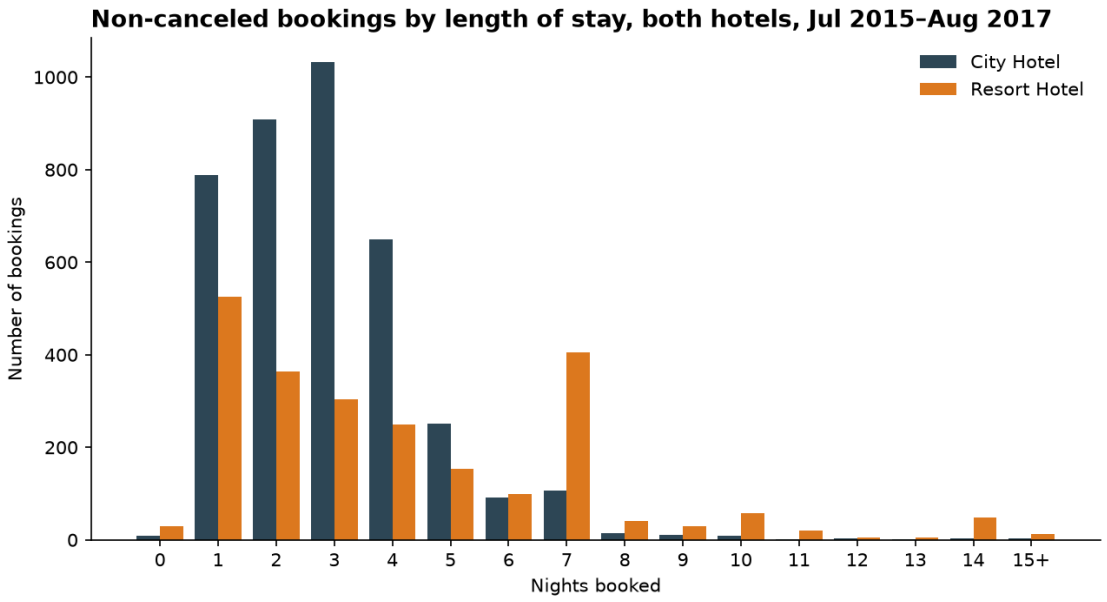


Figure 6. *ADR distribution of non-canceled bookings (adr > 0). n = 6,121 bookings (120 with adr = 0 and 18 missing excluded). Source: outputs/tables/q08_adr_stats_by_hotel.csv*

Financially, when evaluating materialized (non-canceled) bookings with valid pricing (ADR > 0), the two properties operate in distinct economic strata. The median ADR for City Hotel stands at 100.65 EUR, surpassing Resort Hotel (75.00 EUR) by a margin of 25.65 EUR. However, Resort Hotel's interquartile range reveals significantly greater pricing dispersion (Q1-Q3: 51.5-120 EUR), reflecting flexible seasonal pricing architectures inherent to resort models. Note that ADR is strictly a daily rate proxy and excludes macroeconomic seasonal adjustments or ancillary revenue streams.



n = 6,234 non-canceled bookings with nights; 25 missing excluded. 37 zero-night bookings (valid, kept). Source: outputs/tables/q09_stay_nights.csv

Figure 7. *Non-canceled bookings by length of stay. n = 6,234 bookings (25 missing excluded; 37 zero-night valid bookings kept). Source: outputs/tables/q09_stay_nights.csv*

Analyzing the temporal commitment of these 6,234 materialized bookings, 70.29% of City Hotel reservations encompass brief transit stays of 1–3 nights, compared to only 50.72% at Resort Hotel (a delta of 19.6 percentage points). Conversely, Resort Hotel exhibits a pronounced mode at exactly 7 nights (17.3%), with 26.6% of its total volume extending to 7 nights or longer-a stark contrast to City Hotel's mere 3.9%. Analytically, City Hotel functions predominantly as a high-turnover transient node, whereas Resort Hotel primarily caters to sustained weekly leisure cycles.

3.6 Insight 6: Inadequate data sanitation would systematically distort booking volumes, financial metrics, and attrition rates

As detailed previously in **Table 1 (Section 2.1)**, the raw dataset contained 10,325 operational rows; however, rigorous deduplication and entity resolution yielded a precise 10,000 unique bookings, purging 181 absolute duplicates and consolidating 144 fragmented records. Foregoing this preprocessing framework would have severely compromised analytical integrity: a naïve query for 'City Hotel' would overstate volume by 61 instances; the maximum ADR outlier would register an impossible 5,400 EUR instead of the validated 392 EUR; and 41 bookings would misrepresent their final status. Systemically, 3,955 anomalous cells were standardized to NA, marginally inflating the ADR missing rate from 0.24% to 0.36%-a statistically necessary trade-off to eliminate corrupted inputs and ensure the validity of downstream analyses.

4. General Limitations & Recommendations

4.1 General Limitations of the Dataset

1. **Sampling Representativeness:** The data exclusively encompasses two properties in Portugal with arrival dates bridging July 2015 to August 2017. Furthermore, it constitutes a random subset derived from an original repository of 119,390 bookings, intentionally seeded with structural anomalies for pedagogical purposes. Consequently, extrapolating these findings to the broader macroeconomic hospitality industry is empirically invalid.
2. **Absence of Causal Mechanisms:** The analytical framework lacks critical variables such as subjective cancellation rationales, dynamic pricing policies, and competitor benchmarking. Therefore, all comparative analyses of cancellation rates (stratified by lead time, deposit type, market segment, and geography) reflect purely observational correlations. Inferring causality is methodologically unsound due to the profound intersectionality of these variables.
3. **Financial Estimation Constraints:** The adr metric functions strictly as a normalized average daily rate, and revenue_est serves only as a baseline room revenue approximation. These figures omit ancillary service consumption, dining expenditures, and exact cancellation penalty recoveries, precluding their use as definitive total revenue metrics.
4. **Data Sparsity and Granularity:** Only the fiscal year 2016 contains a complete 12-month cycle, restricting year-over-year temporal comparisons. Additionally, the country variable for canceled bookings is susceptible to default-value bias. Certain categorical subsets (e.g., Refundable: 13, Aviation: 21, Room Type H: 23) suffer from extreme sample sparsity, rendering localized statistical inferences highly unstable.

4.2 Data recommendations and follow-up questions

1. **Contextual Cancellation Tracking:** System architecture should be upgraded to capture categorical cancellation rationales and exact cancellation timestamps. This longitudinal data is critical to differentiating between premature and last-minute attrition and is strictly necessary to demystify the anomalous cancellation rates observed within the Non-Refundable deposit cohort.
2. **Comprehensive Revenue Integration:** Future iterations must integrate actualized booking revenue pipelines (encompassing finalized room rates, extracted cancellation fees, and ancillary spend) mapped against a seasonal pricing index. This multidimensional approach will facilitate precise comparisons between theoretical ADR and realized profitability, enabling granular attrition

modeling across intersectional dimensions (Market Segment × Deposit Type × Property).

5. Division

Member	Completed portion	Explicable portion	Contribution rate
Phạm Ngọc Mai	**Data Engineering & Quality Assurance:** Executed Phase A (Data Profiling) and Phase B (Cleaning B1-B8). Developed Phase C (Derived metrics, Monthly summary generation). Ensured 100% compliance with automated testing scripts.	Can comprehensively explain all automated cleaning rules, missing value retention logic, entity resolution, and the data lineage tracking process.	50%
Nguyễn Hà Phương	**Data Analysis & Reporting:** Executed Phase D (10 analytical queries, 8 Matplotlib visualizations). Authored Phase E (Insight formulation, General limitations, Strategic recommendations) Formatted the final documents.	Can comprehensively explain the statistical denominators, filtering constraints, chart configurations, and the business rationale behind the 6 core insights.	50%

Appendix

Appendix A: Extended Statistical Tables

This section provides full cross-tabulations and supplementary data supporting the core insights presented in Section 3.

Table A.1: Reservation Status Distribution by Deposit Type (Detailed)

Note: This table expands upon Insight 3, detailing the exact breakdown of terminal statuses across all deposit categories. Total canceled bookings in Section 3 combine both 'Canceled' and 'No-Show' statuses.

Deposit Type	Total Bookings	Check-Out (Volume / %)	Canceled (Volume / %)	No-Show (Volume / %)	Unknown/ Missing
No Deposit	8,728	6,224 / 71.31%	2,377 / 27.23%	105 / 1.20%	22
Non Refund	1,259	12 / 0.95%	1,244 / 98.81%	0 / 0.00%	3
Refundable	13	8 / 61.54%	5 / 38.46%	0 / 0.00%	0
Total	10,000	6,244 / 62.44%	3,626 / 36.26%	105 / 1.05%	25

Source: outputs/tables/q07_deposit_cancel.csv

Table A.2: Room Reassignment Frequencies by Reserved Room Type

Note: Analyzed exclusively on non-canceled bookings where both reserved and assigned room types are validly recorded (Derived from Query 10).

Reserved Room Type	Total Bookings	Assigned Same Room Type	Reassigned to Different Type (Room Changed)	Reassignment Rate (%)
A	4,287	3,339	948	22.11%
B	64	53	11	17.19%
C	50	46	4	8.00%
D	1,137	1,023	114	10.03%
E	397	350	47	11.84%
Others (F, G, H)	307	287	20	6.51%

Source: outputs/tables/q10_room_change_by_reserved_type.csv

Appendix B: Data Governance & Cleaning Artifacts

This section demonstrates the rigorous data sanitization protocols executed to guarantee analytical integrity, reflecting the automated processes logged in cleaning_log.csv and row_lineage.csv.

Table B.1: Excerpt of the Automated Cleaning Log

Note: The following is a representative sample of the systemic rules applied during Phase B. The full execution log is accessible in the submission package.

Step	Field	Issue Detected	Action Taken	Affected Rows	Check After
B1	booking_id	Whitespaces & lowercase characters	Applied .strip().upper() to standardize key constraints.	215	booking_id format is uniform; exactly 10,145 distinct keys.
B2	reservation_status	Dates misplaced in categorical column	Relocated date strings (YYYY-MM-DD) to reservation_status_date where the destination was empty. Coerced origin cells to NA.	26	No date formats remain in the reservation_status column.
B3	lead_time	Empty cell / missing-value code	Standardized representations (NULL, N/A, NA, ?, -, unknown) strictly to pd.NA without statistical imputation.	10	lead_time: no unrecognized missing-value strings remain.
B4	is_canceled	Contradiction with terminal status	Recomputed binary flag based on valid reservation_status (Check-Out → 0; Canceled/No-Show → 1).	40	100% logical consistency between cancellation flag and status.

Source: outputs/cleaning_log.csv

Table B.2: Resolution of Intra-Record Conflicts during Entity Merging (Phase B8)
During the deduplication of fragmented rows sharing an identical `booking_id`, 144 rows required merging. Among these, 15 bookings exhibited conflicting values in

continuous variables (specifically `adr` and `lead_time`). The resolution heuristic applied was **"First Valid Observation by Ascending row_id"**.

Example of handled conflicts:

- **Booking ID: BK044016** (Merging row R00096): Conflicting `lead_time` values (e.g., 236 vs. 239). Resolved to **236** (associated with the lowest `row_id` representing the earliest data capture).
- **Booking ID: BK049551** (Merging row R01372): Conflicting `adr` values (e.g., 95.5 vs. 95.0). Resolved to **95.5** (associated with the lowest `row_id`).

Methodological Impact: This deterministic rule ensures absolute reproducibility of the merged dataset without synthesizing artificial data (e.g., averaging), preserving the raw integrity of the first recorded transaction.

Source: outputs/row_lineage.csv & Execution environment runtime.